UE24CS342AA2
Data Analytics
Worksheet 4A

Market Basket Analysis & Association Rules

Instructions: Café Aroma recorded 10 customer transactions over one morning. Use the dataset below to answer the questions below. Show all working (raw counts, supports, and confidences) and not just final answers. Unless a question states otherwise, use min_sup = 30% and min_conf = 70%.

Dataset:

Transaction ID	Items Purchased
T1	Coffee, Muffin, Sandwich
T2	Coffee, Muffin
T3	Coffee, Sandwich
T4	Muffin, Sandwich
T5	Coffee, Muffin, Sandwich
T6	Tea, Juice
T7	Coffee, Muffin, Sandwich
T8	Coffee, Tea
T9	Muffin, Juice
T10	Coffee, Muffin

Part A: Mining Frequent Itemsets

Q1. Support Count and Support

Directly scan the transaction table and report:

- (a) σ({Coffee})
- (b) σ({Coffee, Muffin, Sandwich})
- (c) s({Coffee, Muffin, Sandwich})

Q2. Candidate and Frequent 1-itemsets
(a) List C1: every single item with its support count (how many transactions it appears in).

(b) Find L1: keep only the items with count ≥ 3. Which item(s) are dropped?

(c) The Apriori property says we can skip any itemset that contains a dropped item. Why?

Q3. Candidate and Frequent 2-itemsets
(a) Make C2: join L1 with itself to get all possible pairs.

(b) Count each pair in the transaction table. Show the count for every pair, even the ones that get dropped.

(c) Find L2: keep only the pairs with count ≥ 3.

Q4. Candidate and Frequent 3-itemsets
(a) Make C3: build the triple(s) from the pairs in L2.

(b) Count each triple in the transaction table. If its count is ≥ 3, it goes in L3. Otherwise it is dropped.

Part B: Generating Association Rules

Q5. Rule Confidence
The 3-itemset l = {Coffee, Muffin, Sandwich} from Q4 is frequent. Using the lecture formula for a rule s → (l − s), confidence = support(l) / support(s), compute the confidence of these three rules:
(a) {Coffee, Muffin} → {Sandwich}
(b) {Coffee, Sandwich} → {Muffin}
(c) {Muffin, Sandwich} → {Coffee}
Using min_conf = 70%, which of these rules are strong rules and which are filtered out?

Q6. Is a Strong Rule Always Useful? (Bonus)
Additional information (not shown in the table above):
Café Aroma also tracked newspaper purchases for the same 10 transactions. Of the 10 transactions, 8 included a newspaper. Of the 7 transactions that included coffee, 5 also included a newspaper.
(a) Compute the confidence of the rule Coffee → Newspaper. Does it meet min_conf = 70%?
(b) Compute the overall support of Newspaper, s(Newspaper), ignoring coffee. Compare it with your answer to (a) and say in one or two sentences whether this rule is useful.